PROJECT

1. Hilbert scheme of points

Throughout X will be a projective complex surface. The formal construction of the Hilbert scheme is omitted. The essential property of the Hilbert scheme is that a point (over $\mathbb{C}$) of $X^{[n]}$ is a 0-dimensional closed subscheme of X of length n. We think of points of $X^{[n]}$ interchangeably as collections of n points of X and ideal sheaves.

Remark 1.1. There is some concern about considering the $\mathbb{C}$-points. I think this means that the residue field i.e. the stalk over its maximal ideal, should be $\mathbb{C}$ on that point. But also in Video 9 Anthony makes the remark that he will call $V(\mathbb{C})$ just V.

The following construction distinguishes the Hilbert scheme from the symmetric power S^nX of the surface X. In some cases this is an isomorphism, but not always.

The *symmetric power* of X is

$$S^nX = \underbrace{X \times \ldots \times X}_{n \text{ copies}} / S_n$$

where the symmetric group S_n acts by permutating copies. Thus, a point of S^nX is an S_n-orbit on the cartesian product $X \times \ldots \times X$, that is, an unordered n-tuple of points of X, with allowed repetition. In algebraic geometry these are called "algebraic cycles" and are denoted as formal sums $\sum_{x\in X} n_x[x]$ where $\sum_{x\in X} n_x = n$, with $n_x \in \mathbb{N}$ and $[x]$ are formal symbols.

If Z is a 0-dimensional closed subscheme of X, the *length* of Z is $\dim \Gamma(\mathcal{O}_Z)$. Any such Z is supported at finitely many points. We have that $\dim \Gamma(\mathcal{O}_Z) = \sum_{x\in \operatorname{supp} Z} \Gamma(\mathcal{O}_{Z,x})$.

This gives a map

$$\begin{aligned} \pi : X^{[n]} &\longrightarrow S^nX \\ Z &\longmapsto \underbrace{\sum_{x\in \operatorname{supp} Z} \dim \Gamma(\mathcal{O}_{Z,x} x)}_{\text{cycle of } Z} \end{aligned}$$

called the *Hilbert-Chow morphism*.

Let us have short digression on the concept of length

The stalk is a colimit in the category of commutative rings, which makes it an $S = k[x_0, \ldots, x_n]$-module Here's Stacks Project Definition 10.52.1 of length of a module:

Definition 1.2. Let R be a ring. For any R-module M we define the *length* of M over R by the formula

$$\operatorname{length}_R(M) = \sup\{n \mid \exists\ 0 = M_0 \subset M_1 \subset \ldots \subset M_n = M,\ M_i \neq M_{i+1}\}.$$

But we also have Lemma 10.52.5:

Lemma 1.3. *Let R be a ring with maximal ideal $\mathfrak{m}$. Suppose that M is an R-module with $\mathfrak{m}M = 0$. Then the length of M as an R-module agrees with the dimension of M as a $R/\mathfrak{m}$ vector space. The length is finite if and only if M is a finite R-module.*

So basically this is what we mean by $\dim \Gamma(\mathcal{O}_{Z,x})\ldots$ But no… unfortunately I'm not done here: because it's the dimension of the *sections*, not the dimension of the module itself. So perhaps this can be sorted by putting $\Gamma(\mathcal{O}_Z) \simeq H^0(\mathcal{O}_Z)\ldots$ Anyway this is just some nuance.

Remark 1.4. In the case of X being a smooth quasiprojective surface we don't need to considered the reduced version of $X^{[n]}$; it's a theorem that in this case it is already reduced.

Anyway, the point probably is that when a scheme somehow counts two points instead of one, the height of the stalk at that point will be 2. I guess. This distinguishes $X^{[n]}$ from $S^n X$.

- Search for a variety $(\mathbb{A}^2)^{[n]}$ whose points are in bijection with $\{I \overset{\text{normal}}{<} \mathbb{C}[z_1, z_2] : \dim_{\mathbb{C}} /I\}$.
- We found such a variety, constructed as
$$H := \tilde{H}/\mathrm{GL}_n(\mathbb{C})$$
where
$$\begin{aligned}\tilde{H} = \{&(B_1, B_2, v) \in \mathrm{Mat}_n(\mathbb{C}) \oplus \mathrm{Mat}_n(\mathbb{C}) \oplus \mathbb{C}^n \\ &: [B_1, B_2] = 0 \text{ and } {\scriptstyle v \text{ generates } \mathbb{C}^n \text{ under} \atop \scriptstyle \text{the action of } B_1 \text{ and } B_2}\}\end{aligned}$$
The latter condition is called the *stability condition*. This is the intersection of a closed set and an open subset, so it's a quasi-affine variety.
- $\mathrm{GL}_n(\mathbb{C})$ acts freely on $\tilde{H}$ by
$$g \cdot (B_1, B_2, v) = (gB_1g^{-1}, gB_2g^{-1}, gv).$$
so we can take the quotient, called H.
- Then H corresponds to codimension-n ideals of $\mathbb{C}[z_1, z_2]$. This is because the points of $\tilde{H}$ [...].
- Compare with the construction of the Grassmanian.
- **Lecture 2:** prove that H is smooth.
- **Lecture 3:** prove that functor is representable.

2. Heisenberg aglebra

Recall our definition of the basic Heisenberg Lie algebra: $\hat{\mathfrak{a}} = \bigoplus_{n\in\mathbb{Z}} \mathbb{C}h_n \bigoplus \mathbb{C}_K$ with bracket given by $[h_m, h_n] = m\delta_{m,-n}K$ and $[K, \hat{\mathfrak{a}}] = 0$.

Before continuing let me explain Emily's construction of the Heisenberg algebra in Video 8. Essentially she constructs the affine Lie algebra associated to a commutative Lie algebra (a lattice) and defines Heisenberg as the central extension of that. Though our construction of affine Lie algebras allowed for nonzero brackets in the original Lie algebra, in our definition of Heisenberg there's also no appearence of the bracket of the original Lie algebra: instead we take the underlying Lie algebra to be 1-dimensional!

Let Γ be a lattice (i.e. a finitely generated torsion-free abelian group) equipped with a positive definite symmetric bilinear form $(\cdot,\cdot) : \Gamma \times \Gamma \to \mathbb{Z}$. E.g. $\Gamma = \sqrt{N}\mathbb{Z}$, $(\sqrt{N}n, \sqrt{N}m) = Nnm$.

Turn Γ into a commutative Lie algebra $\mathfrak{g} := \Gamma \otimes_{\mathbb{Z}} \mathbb{C}$ by putting zero bracket. Now consider the *loop algebra* $\mathfrak{g}[[t]]$, which is "spanned topologically" by $\alpha \otimes t^n$, $\alpha \in \Gamma$ and $n \in \mathbb{Z}$ (later in the video it is explained that this is the same as our loop algebra $L\mathfrak{a} = \mathfrak{a}[t, t^{-1}] = \mathfrak{a} \otimes_{\mathbb{C}} \mathbb{C}[t, t^{-1}]$). Notice that $(\cdot,\cdot)$ extends to $\mathfrak{h}$.

Definition 2.1. The *Heisenberg Lie algebra* $\hat{\mathfrak{a}}_\Gamma$ associated to $(\Gamma, (\cdot,\cdot))$ is the central extension

$$0 \longrightarrow \mathbb{C}K \longrightarrow \hat{\mathfrak{a}}_L \longrightarrow \mathfrak{a}[[t]] \longrightarrow 0$$

with Lie bracket given by

- $[K, \hat{\mathfrak{a}}] = 0$, (this is why it is called *central* extension; K commutes with everything),
- and
$$[\alpha \otimes f(t), \beta \otimes g(t)]_{\hat{\mathfrak{a}}_L} := -(\alpha,\beta)(\mathrm{Res}_{t=0} f dg)K.$$

One checks that in fact this bracket satisfies the definition of Lie bracket, and that in monomials it acts by:

$$[\alpha \otimes t^m, \beta \otimes t^n] = -(\alpha,\beta) n \delta_{m,n-n} K.$$

From now on we denote $\alpha(n) := \alpha \otimes t^n$ for $\alpha \in \Gamma$.

Now we make some remarks on the universal enveloping algebra of Heisenberg. Recall that an explicit construction of the universal enveloping algebra of a Lie algebra is given by a quotient of the tensor algebra by an ideal generated by the commutation relations. Emily claims that the universal enveloping algebra of Heisenberg $U(\hat{\mathfrak{a}})$ contains finite linear combinations of finite products of possible *infinite* sums of $\alpha(n)$ and K. The "topological completion" of $U(\hat{\mathfrak{a}})$, denoted $\tilde{U}(\hat{\mathfrak{a}})$, allows from infinite sums. So, this topological completion is spanned by infinite linear combinations of finite products of the $\alpha(n)$ and K.

In this case there is a PBW theorem that says if $\alpha_1, \ldots, \alpha_r$ is a basis of Γ then $\tilde{U}(\hat{\mathfrak{a}})$ is "topologically spanned" by monomials of the form $\alpha_1(n_1^1)\alpha_1(n_1^2)\ldots\alpha_1(n_1^{k_1})\alpha_2(n_2^1)\ldots$, where $n_1^1 \leq n_1^2 \leq \ldots \leq n_1^{k_1}$. That is, we have monomials in infinitely many variables.

Like in the finite-dimensional case, any (continuous) $\hat{\mathfrak{a}}$-module is equivalent to a (continuous) $\tilde{U}(\hat{\mathfrak{a}})$-module. (Cf. [Kac10, Corollary 22.1]: any representation $\pi : \mathfrak{g} \to \mathrm{End}(V)$ of extends uniquely to a homomorphism of associative algebras $U(\mathfrak{g}) \to \mathrm{End}(V)$.)

Next consider the following characterization of irreducible $\hat{\mathfrak{a}}$-modules. Let V be an irreiducible $\hat{\mathfrak{a}}$-module. Suppose that $K \in \mathrm{End}(V)$ has an eigenvalue (this will happen if we impose certain conditions on V like countable dimension, see Dixmier's lemma). Since K is central in $\hat{\mathfrak{a}}$, the endomorphism K is a morphism of $\hat{\mathfrak{a}}$-modules (indeed: a morphism of representations is just a $\hat{\mathfrak{a}}$-linear map).

Now choose an eigenvector v_0 with eigenvalue κ. Then the map $K - \kappa\mathrm{id}$ is also a morphism of $\hat{\mathfrak{a}}$-modules with nonzero kernel ($v_0 \in \mathrm{Ker}$). By Schur's Lemma, which says that any map of irreducible modules is either zero or on an isomorphism, this map must be zero.

In conclusion, in any irredicible $\hat{\mathfrak{a}}$-module, K acts by multiplication by some complex number κ which we call the *level*. Often we just study representation of level κ (not worrying any more about the restrictions of V and barely asking that K acts by multiplication by κ).

Since $\hat{\mathfrak{a}}$-modules are equivalent to $\widetilde{U}(\hat{\mathfrak{a}})$-modules, it turns out that $\hat{\mathfrak{a}}$-modules of level κ are equivalent to modules over the quotient $\widetilde{\mathcal{H}}$ of the universal enveloping algebra by the ideal generated by $K - \kappa\mathrm{Id}$. Furthermore, any choice of $\kappa \neq 0$ gives an equivalent representation theory. Indeed, rescaling $\alpha(n) \mapsto \kappa^{1/2}\alpha(n)$ gives an isomorphism of representations. Notice this is not the case for any affine Lie algebra.

Now we turn to the Fock representation. In view of the construction so far we might as well consider level-1 representations, that is, K acts as the identity (as we did in our course from the very beginning!).

For a complex number $\mu \in \mathbb{C}$ define a highest weight module

$$F^\mu = U(\hat{\mathfrak{a}}) \otimes_{U(\hat{\mathfrak{a}}_+)} \mathbb{C}|\mu\rangle.$$

Then $h_0|\mu\rangle = \mu|\mu\rangle$ ($\mu \in \mathbb{C}$). (The Heisenberg vertex algebra is F^0, i.e. $\mu = 0$.)

It turns out that F^μ is irreducible. This is a special feature of $\hat{\mathfrak{a}}$, proved using Shapovalov form. For more complicated Lie algebras $\hat{\mathfrak{g}}$, Verma $\hat{\mathfrak{g}}$-modules can be non-irreducible.

Also, F^μ satisfies the universal property that if V is an $\hat{\mathfrak{a}}$-module with a vector $v \in V$ such that

$$h_0 v = \mu v \qquad \text{and} \qquad h_{>0} v = 0$$

then there exists a morphism of $\hat{\mathfrak{a}}$-modules

$$f : F^\mu \longrightarrow V$$
$$\text{such that } |\mu\rangle \longmapsto v.$$

Now, if $v \neq 0$, so $f(F^\mu) \neq 0$, then $f(F^\mu) \cong F^\mu / \operatorname{Ker}(f)$. But since F^μ is irreducible, $\operatorname{Ker}(f) = \{0\}$, i.e. $f : F^\mu \to V$ is an inclusion.

If, furthermore, V is generated by V, i.e. $U(\hat{\mathfrak{a}}) \cdot v = V$, then $V = f(F^\mu) \simeq F^\mu$.

As corollary, any $\hat{\mathfrak{a}}$-module generated by a highest weight vector v (such that $h_0 v = \mu v$) is isomorphic to F^μ.

Let's now focus on the "basic" Heisenberg Lie algebra, i.e. we fix $\mu = 0$ from now on. Then $F^0 \simeq \mathbb{C}[x_1, x_2, \ldots]$. Define the *energy* by

$$\Delta(x_1^{k_1} x_2^{k_2} \ldots x_s^{k_s}) = \sum_j k_j.$$

I think the is basically the grading we put on F^0. The natural grading on $\mathbb{C}[x_1, \ldots, x_n]$ is degree, so that the factors of the decomposition are generated by homogeneous monomials. Here is not degree but this "weighted" sum.

Then the character is

$$\sum_{\Delta=0}^{\infty} q^\Delta \dim(F^0_\Delta) = \prod_{m=1}^{\infty} \frac{1}{1-q^m}.$$

Consider the following similar construction. Let $\mathfrak{h} = \mathbb{C}^2 = \mathbb{C}x + \mathbb{C}y$ and put the symmetric bilinear form given by $(x,x) = (y,y) = 1$ and (x,y) (i.e. with Gram matrix the identity). Equip it with a null bracket to make it an abelian Lie algebra.

Then $\hat{\mathfrak{h}} = \mathfrak{h} \otimes \mathbb{C}[[t^{\pm}]] \oplus \mathbb{C}K$ has bracket $[at^m, bt^n] = \underbrace{[a,b]}_{=0} + m\delta_{m,-n}(a,b)K$. The Fock space/Verma module for $\hat{\mathfrak{h}}$ is $F^0 = \mathbb{C}[x_1, y_1, x_2, y_2, \ldots]$, and its character is

$$\sum q^{\Delta} \dim F^0_{\Delta} = \prod_{m=1}^{\infty} \frac{1}{(1-q^m)^2}.$$

In the quotient, one factor comes from the $x_1, x_2, \ldots$ and the other one from $y_1, y_2, \ldots$. Actually we can spell this out further. Notice that F^0 is spanned by monomials

$$x_1^{k_1} x_2^{k_2} \ldots x_2^{k_s} y_1^{\ell_1} y_2^{\ell_2} \ldots y_t^{\ell_t}.$$

Each of this monomials corresponds to exactly one term in the product

$$x \begin{cases} m=1 & (1 + \underline{q} + q^2 + q^3 + \ldots)\times \\ m=2 & (1 + q^2 + \underline{q^4} + q^6 + \ldots)\times \\ m=3 & (\underline{1} + q^3 + q^6 + q^9 + \ldots)\times \end{cases}$$

$$y \begin{cases} m=1 & (\underline{1} + q + q^2 + q^3 + \ldots)\times \\ m=2 & (1 + q^2 + q^4 + \underline{q^6} + \ldots)\times \\ m=3 & (\underline{1} + q^3 + q^6 + q^9 + \ldots)\times . \end{cases}$$

For instance, the monomial $x_1 x_2^2 y_2^3$ corresponds to the underlined term $(q)(q^4)(1)(1)(q^6)(1)\ldots$.

Digression/unexplanation. It is indeed a nice picture, but do notice that's at least for me now it's now obvious why it's true. The definition of the character as $\sum_{\Delta=0}^{\infty} q^{\Delta} \dim(F^0_{\Delta})$, so q is a formal variable storing the data of the dimensions of each factor in the Δ-decomposition of F^0. That is, there is a collection of size $\dim F^0_{\Delta}$ of monomials for every energy $\Delta = n$. So pick all the monomials that give $\Delta = 3$. They are found in the sea of q above. So you can factor q^3 out of all of them and you'll get a number, the coefficient. So I guess you factor $\Delta = 1$ first, then $\Delta = 2$ and so on untill you arrive at $\sum_{\Delta=0}^{\infty} q^{\Delta} \dim(F^0_{\Delta})$. **Edit.** In fact, we did prove this, once at least, using that the Heisenberg algebra is the charge-0 part of the fermion algebra, i.e. $F^{(0)}$.

This construction generalizes for $U \cong \mathbb{C}^n$ with a symmetric non-degenerate bilinear form $(\cdot,\cdot) : U \times U \to \mathbb{C}$. There exists a basis $x^1, \ldots, x^n$ of U such that $(\cdot,\cdot)$ has Gram matrix the identity. Then the Fock module F^0 is spanned by monomials with are complicated to write or read, but the character is actually simple:

$$\prod_{m=1}^{\infty} \frac{1}{(1-q^m)^n}.$$

It's possible to keep track of the total number of x and y, as well as the total energy.

Recall from lectures that we found the character of the charged Fermions is

$$\chi_{F^{(m)}}(q) = \sum_{\Delta} \dim F^{(m)}_{\Delta} q^{\Delta} = q^{m/2} \prod_{k=1}^{\infty} \frac{1}{1-q^k}.$$

and that in fact the charge-c component in the (charge,energy)-decomposition of F^{ch} is in fact isomorphic to the Fock space $F^c = U(\hat{\mathfrak{a}}) \otimes_{U(\hat{\mathfrak{a}}_+)} \mathbb{C}|c\rangle$.

3. Symplectic quotient

The coadjoint action induces orbits in $\mathfrak{g}^*$. The tangent space at any point of a coadjoint orbit is given by infinitesimal vector fields, that is,

$$T_\xi \mathcal{O} = \{u_{\mathfrak{g}^*}(\xi) : \forall u \in \mathfrak{g}\}.$$

We may introduce a symlpectic form $\omega \in \Lambda^2 T^*_\xi \mathcal{O}$ on the tangent space at some point of a coadjoint orbit by

$$\omega_\xi(X, Y) := \xi([u, v]) = -u_{\mathfrak{g}^*}(\xi)v = v_{\mathfrak{g}^*}(\xi)(u)$$

which does not depend on $u, v \in \mathfrak{g}$.

Theorem 3.1 (KKS). *$\xi \mapsto \omega_\xi$ is symplectic on $\mathcal{O}$.*

Proof. Non degeneracy is easy. Then show that ω is G-invariant. Closedness follows from Jacobi identity. $\square$

4. Correspondences

If $f : X \to Y$ is continuous, we have an induced map $f_* : H_i(X) \to H_i(Y)$. If a group G acts on X, then it acts on $H_\bullet(X)$ (this is a way producing a representation, it follows from the functoriality of homology).

Thinking of the graph f as a subset of $X \times Y$, we can ask if a closed subset $Z \subset X \times Y$ (not necessarily the graph of a continuous map) will induce a map on homology $H_i(X) \to H_i(Y)$.

The natural idea is to produce a map using the projections

$$\begin{tikzcd} & Z \arrow[dl, "p_X"'] \arrow[dr, "p_Y"] & \\ X & & Y \end{tikzcd}$$

and essentially pulling back a class and then pushing it forward, i.e.

$$H_i(X) \xrightarrow{p_X^*} H_i(Z) \xrightarrow{p_{Y,*}} H_i(Y)$$

but it turns out that a functor cannot be covariant and contravariant at the same time, that is, the pushforward and pullback cannot be defined for all maps. So we shall give up functoriality and define the pushforward and pullback on some classes of maps instead.

Now we construct Borel-Moore homology, also known as locally finite homology. We always consider homology with $\mathbb{C}$ coefficients.

By *manifold* we mean a connected, oriented finite dimensional manifold over $\mathbb{R}$. The primary examples of manifolds will be smooth complex varieties. For example, V a smooth irreducible variety of dimension d over $\mathbb{C}$, which is a manifold of dimension $2d$, with the canonical orientation coming from the complex structure.

By *space* we mean a topological space that is a proper deformation retract of a closed neighbourhood of a manifold. For example, a variety V over $\mathbb{C}$. Also projective varieties are spaces seen as submanifolds of $\mathbb{P}^n$.

Let X be a space. Recall that simplicial homology is the homology of the singular chain complex

$$\cdots \longrightarrow C_i(X) \xrightarrow{\partial} C_{i-1}(X) \longrightarrow \cdots$$

where $C_i(X)$ is the set of finite linear combinations of i-simplices in X.

For Borel-Moore homology $H_\bullet^{BM}(X)$ we use $C_i^{lf}(X)$, which is the set of locally finite linear combinations of singular simplices. Here locally finite means that every point $x \in X$ is in a neighbourhood that intersects only finitely many simplices.

Borel-Moore homology is not homotopy invariant. To see this recall that the homology groups of $\mathbb{R}^2$ are $H_1(\mathbb{R}^2) = \mathbb{C}$ and $H_i(\mathbb{R}^2) = 0$ for $i \neq 1$. However, $H_2^{BM}(\mathbb{R}^2) = \mathbb{C}$ since a tiling of triangles is locally finite chain of 2 simplices, has no boundary (since the boundaries of all triangles cancel each other), and is clearly not the boundary of a 3-simplex; thus this tiling is a nonzero element of $H_2^{BM}(\mathbb{R}^2)$. Further $H_0^{BM}(\mathbb{R}^2) = 0$ since ray of 1-simplices is a locally finite chain whose boundary is a ponit; thus a point is a boundary and thus has trivial homology class.

Facts:

- If X is compact, $H_i^{BM}(X) \cong H_i(X)$.
- If X is not compact, $H_i^{BM}(X) \cong H_i(\hat{X}, \infty)$ where $\hat{X}$ is the one-point compactification of X and $H_i(\hat{X}, \infty)$ is the relative homology group.
- If the dimension of the manifold where X is embedded is m, there is a canonical isomorphism $H_i^{BM}(X) \cong H^{m-i}(M, M \setminus X)$. In other words, there is a perfect pairing
$$H_i^{BM}(X) \times H_{m-i}(M, M \setminus X) \to \mathbb{C}$$
given by "intersection" (where the quotes account for some transversal intersection condition that we would require before computing the intersection of two simplicial chains).

In the previous example of the tiling of $\mathbb{R}^2$ by triangles, we can take $X = M = \mathbb{R}^2$. To see how the intersection pairing works just intersect the 0-chain in singular homology with the tiling in Borel-Moore homology: since the point lies in exactly one triangle (after suitable perturbation) their intersection is 1. Then the pairing gives 1 between those two chains. Notice this wouldn't work singular homology: we need the infinite assumption on the 2-chain to make sure that the whole plain is tiled by triangles and we cannot move the point to a region where it is not contained in any triangle.

Key properties of Borel-Moore homology (see [**?**, Section 4.2]):

(1) If $f : X \to Y$ is proper (e.g. inverse image of compact is compact) then $f_* : H_i^{BM}(X) \to H_i^{BM}(Y)$. This is functorial in proper maps, i.e. preserves compositions. This is because the inverse image of a locally finite intersection with a simplicial chain remains locally finite with respect to the preimage chain by properness.

(2) (Pushforward.) Let $U \subset X$ be open, $F = X \setminus U$, $u : U \hookrightarrow X$ and $f : F \hookrightarrow X$ inclusions. Then $u^* : H_i^{BM}(X) \to H_i^{BM}(U)$ fitting into a long exact sequence
$$\cdots \longrightarrow H_i^{BM}(F) \xrightarrow{f_*} H_i^{BM}(X) \xrightarrow{u^*} H_i^{BM}(U) \xrightarrow{\partial} H_{i-1}^{BM}(F) \longrightarrow \cdots$$
This can be constructed from the isomorphism above with singular cohomology.

(3) (Pullback.) If $f : X \to Y$ is a locally trivial fibration with fiber a manifold of dimension m (over $\mathbb{R}$), then $f^* : H_i^{BM}(Y) \to H_{i+m}^{BM}(X)$.

(4) (Intersection pairing.) If X and Y are both nicely embedded in the same manifold M,

$$\cap_M : H_i^{BM}(X) \times H_j^{BM}(Y) \to H_{i+j-m}^{BM}(X \cap Y).$$

The proof of this product is by using the isomorphism with relative cohomology, and then using usual cap product.

(5) (Fundamental classes.)
- If V is a smooth irreducible variety of dimension d over $\mathbb{C}$ then let $[V]$ for the canonical generator of $H_{2d}^{BM}(V) \cong H^0(V) \cong \mathbb{C}$. (Where the last isomorphism is just evaluating the 0-cochain and evalating in the only point that exists, up to homology, since V is connected.) This is the *fundamental class* of V.
- If V is an irreducible variety of dimension d over $\mathbb{C}$, let $[V] \in B_{2d}^{BM}(V)$ be the unique d such that $u^*[V] = [V_{\text{reg}}]$. That is, $[V]$ is the class corresponding to the pullback of u on the open set of regular points of V. The pullback in this degree is injective and thus this class is unique.
- For V not necessarily irreducible of dimension d, then $H_{2d}^{BM}(V)$ has basis $[V_1], \ldots, [V_k]$ where V_i are the d-dimensional irreducible components of V. In this setting, the fundamental class is defined as the sum of the fundamental classes of those components, i.e. $[V] = [V_1] + \ldots + [V_k]$.

A special feature of BM homology is that we can define these fundamental classes. In particular, the fundamental class of the whole space, as in our example with $\mathbb{R}^2$, is a BM-class because we admit infinite chains.

Finally we arrive at correspondences.

Situation 4.1. Let X_1, X_2 be smooth irreducible varieties of dimensions d_1, d_2. Let Z be a d-dimensional closed subvariety of $X_1 \times X_2$ such that the projections p_1 and p_2 are proper.

Our objective is to define a correspondence from the following diagram.

$$\begin{tikzcd} & Z \arrow[dl, "p_1"'] \arrow[dr, "p_2"] & \\ X_1 & & X_2. \end{tikzcd}$$

The correspondence is

$$c_{[Z]} : H_i^{BM}(X_2) \to H_{i+2d-2d_2}^{BM}(X_1),$$

defined by first applying the pullback property

$$H_i^{BM}(X) \xrightarrow{p_2^*} H_{i+2d_1}^{BM}(X_1 \times X_2)$$

because the projection onto the second factor is a locally trivial (in fact trivial) fibration with fibers of real dimension $2d_1$. So by **??** we obtain the dimension $i + 2d_1$.

Them we take the intersection pairing with the fundamental class of Z, which lives in the $2d$-th BM homology of Z. According to 4 this should be a map

$$H^{BM}_{i+2d_1}(X_1 \times X_2) \xrightarrow{\cap_{X_1\times X_2}[Z]} H_{i+2d_1+2d-\underbrace{(2d_1+2d_2)}_{\substack{\text{real dimension}\\ \text{of ambient space}}}}(Z).$$

Finally take the pushforward map of p_1, which we assumed to be proper. All in all, using pullback, intersection and pushforward we have constructed

$$H^{BM}_i(X) \xrightarrow{p_2^*} H^{BM}_{i+2d_1}(X_1 \times X_2) \xrightarrow{\cap_{X_1\times X_2}[Z]} H_{i+2d_1+2d-(2d_1+2d_2)}(Z) \xrightarrow{(p_1)_*} H_{i+2d-2d_2}(X_1).$$

Morally we think of the first two steps as just doing pullback from X_2 to Z, but this can't be done so simply since we don't want to assume that p_2 is locally trivial (this won't be true in practice), and that's the only case when we get pullbacks. So that's why we pullback to a larger space where Z is contained and then take intersection.

Remark 4.2. (1) We can also define correspondences $c_\alpha : H^{BM}_1(X_2) \to H^{BM}_{i+2d-2d_2}(X_1)$ for any $\alpha \in H^{BM}_{2d}$.
(2) If Z is the graph of a proper morphism $f : X_2 \to X_1$, then $c_{[Z]} = f_*$. So that in fact correspondences generalize the usual pushforward notion we have for BM homology.

Next we show how to compose correspondences. Consider the diagram:

$$\begin{tikzcd}
 & & Z_{12}\times_{X_2} Z_{23} \arrow[dl] \arrow[dr] & & \\
 & Z_{12} \arrow[dl] \arrow[dr] & & Z_{23} \arrow[dl] \arrow[dr] & \\
X_1 & & X_2 & & X_3
\end{tikzcd}$$

Then the composition of the correspondences in the bottom triangles is the correspondence of the outer triangle. Here we think of the fiber product as

$$\begin{aligned} Z_{12} \times_{X_2} Z_{23} &= \{(x_1,x_2,x_3) \in X_1 \times X_2 \times X_3 : (x_1,x_2) \in Z_{12}, (x_2,x_3) \in Z_{23}\} \\ &= p_{12}^{-1}(Z_{12}) \cap p_{23}^{-1}(Z_{23}), \end{aligned}$$

where p_{12} is the projection from the triple product to $X_1 \times X_2$ and p_{23} is similarly defined.

We get

$$Z_{12} \circ Z_{23} = p_{13}(p_{12}^{-1}(Z_{12}) \cap p_{23}^{-1}(Z_{23})$$

so that

$$Z_{12} \circ Z_{23} = \{(x_1,x_3) \in X_1 \times X_3 : \exists x_2 \in X_2 \text{ s.t. } \substack{(x_1,x_2)\in Z_{12} \\ (x_2,x_3)\in Z_{23}}\}.$$

So that if Z_{12} and Z_{23} where the graphs of two functions, then $Z_{12} \circ Z_{23}$ would be the graph of their composition. Based on this observation, it would be nice if

$$c_{[Z_{12}]} \circ c_{[Z_{23}]} = c_{[Z_{12}\circ Z_{23}]},$$

which happens in the case of graphs, but this is false in general. Tracing through the definitions, we see it's one of these more general correspondences that is associated to a class that is not the fundamental thing of a single thing. So instead of taking the fundamental class of $Z_{12} \circ Z_{23}$, we get

$$c_{[Z_{12}]} \circ c_{[Z_{23}]} = C_{(p_{13})_*(p_{12}^*[Z_{12}] \cap_{X_1 \times X_2 \times X_3} p_{23}^*[Z_{23}]},$$

which is an element of $H^{BM}_{2\dim Z_{12} + 2\dim Z_{23} - 2\dim X_2}(Z_{12} \circ Z_{23})$ as one can check.

Example 4.3. (See the second part of video 9.) As an example of this construction, we may define an $\mathfrak{sl}_2$-action on $\bigoplus_{a,i} H_i^{BM}(T^*\mathrm{Gr}(a, N))$. where $\mathrm{Gr}(a, N)$ is the Grassmanian. Writing $\mathfrak{sl}_2 = \mathbb{C}\{e, h, f\}$, the idea is to construct the maps representing e and f as correspondences. The task is to define a subvariety Z of the product of the appropriate cotangent bundles, and prove that the associated correspondences satisfy the commutativity relations that they should. Further, this representation is irreducible since it happens have 1-dimensional weight spaces, which characterizes the only irreducible representation of $\mathfrak{sl}_2$.

5. Main construction

We want to produce a representation π of the Heisenberg (super)algebra $\mathfrak{h}$ on $V = H_*(X)$. The key idea is to consider some subvarieties of direct products and use their fundamental classes to produce operators on the homology. The main tool for this are "correspondences".

Let $n \geq i > 0$. Define $P[i]$ as a subset of $\prod_n X^{[n-i]} \times X^{[n]} \times X$ by

$$P[i] = \prod_n \{ (\underbrace{\mathcal{I}_1}_{\in X^{[n-i]}} \supseteq \underbrace{\mathcal{I}_2}_{\in X^{[n]}}, \underbrace{x}_{\in X}) : \mathrm{supp}(\mathcal{I}_1/\mathcal{I}_2) = \{x\}\}.$$

(For $i < 0$, we swap $\mathcal{I}_1$ and $\mathcal{I}_2$ in this definition.)

We have projections

$$\begin{tikzcd} & \prod_n X^{[n-i]} \times X^{[n]} \times X \arrow[dl, "\Pi_i"'] \arrow[dr, "\varpi_i"] & \\ X & & \prod_n X^{[n-i]} \times X^{[n]} \end{tikzcd}$$

Proposition 5.1. $\dim_{\mathbb{C}} P[i] \cap (X^{[n-i]} \times X^{[n]}) = 2n - i + 1$.

Next we define correspondences. For $i > 0$, let $\alpha \in H_*^{BM}(X)$, $\beta \in H_*(X)$ and consider

$$P_\alpha[i] = \varpi_*(\Pi^*\alpha \cap [P[i]])$$
$$P_\beta[i] = \varpi_*(\Pi^*\beta \cap [P[-i]]).$$

Notice that it's not clear why we choose α and β in different kinds of homology. At this point we may just assume that X is projective so that BM homology coincides with usual homology.

These classes belong to $\prod_n H_*(X^{[n\mp i]}) \otimes H_*(X^{[n]})$. Here the $\mp$ means that if $i > 0$ we are substracting, and if $i < 0$ we are adding.

Recall how correspondences work: if M_1, M_2 are varieties of dimensions d_1, d_2, and $Z \subseteq M_1 \times M_2$ is a variety of dimension d. We have the projections

$$\begin{tikzcd} & Z \subseteq M_1 \times M_2 \arrow[dl, "p_1"'] \arrow[dr, "p_2"] & \\ M_1 & & M_2. \end{tikzcd}$$

Then we have correspondences defined by

$$\begin{aligned} [Z] : H_*(M_2) &\longrightarrow H_*(M_1) \\ c &\longmapsto \pi_{1,*}(p_2^* c \cap [z]), \end{aligned}$$

which is to be read as: take a class in $H_*(M_2)$, pull it back to the homology of $M_1 \times M_2$, cap it with the fundamental class of Z, and push it down to the homology of M_1.

In this way $P_\alpha[i]$ and $P_\beta[-i]$ give maps

$$\bigoplus_n H_*(X^{[n]}) \to \bigoplus_n H_*(X^{[n \mp i]}).$$

via the perfect pairing property.

Theorem 5.2 (Nakajima, Grojnowski)**.** *Suppose* $(deg\alpha)(deg\beta) \equiv 0(mod 2)$. *Then*

$$[P_\alpha[i], P_\beta[j]] = (-1)^{i-1} i \delta_{i,-j} \langle \alpha, \beta \rangle \, Id.$$

In particular, if we assume $V = H_*(X)$ *is concentrated in even degrees, then* $\bigoplus_n H_*(X^{[n]})$ *is a representation of* $\mathfrak{h}_V$ *with highest weight vector the generator of* $H_0(X^{[0]}) \cong \mathbb{Q}$.

It is not clear from this that this representation is irreducible, but that is in fact true.

Theorem 5.3 (Nakajima)**.** *This representation is irreducible by considering graded dimensions, which we will not discuss.*

This about half of video 10. In the rest of the video an outline of the proof is explained. I didn't go over this yet.

6. Götsche's formula

So far I didn't investigate how this formula can be derived from Theorem 5.2. This is not explained in the videos but most likely it is in [Nak].

For any variety Z (or manifold) we can form the *Poincaré polynomial* of Z, which is the generating function of the Betti numbers of Z

$$P_t(Z) = \sum_{m \geq 0} b_m(Z) t^m, \qquad b_m(Z) = \mathrm{rank} H_m(Z).$$

The idea is to consider the Poincaré polynomials for each each of the n-th Hilbert schemes and look at a generating function of all these polynomials That is, this is some sort of "bi-generating function in t and n":

$$\sum_{n \geq 0} q^n P_t(X^{[n]}) = \text{[Nak95, Equation 1.1]}.$$

7. Worked expample

Fix $x \in X$ and let $\alpha = [X]$ is the GH class of X, and $\beta = [x]$ the class of the little point x. Take a set of distinct n points $x_1, \dots, x_n \in X \setminus \{x\}$ non of which is x. Identify them with the corresponding ideal $\mathcal{I}$. We are going to intercheangbly refer to those points as points and as $\mathcal{I}$.

Let's calculate

$$[P_\alpha[1], P_\beta[-1]][\mathcal{I}].$$

If you look back at the relation, we want this to give $\mathcal{I}$.

References

[Kac10] Victor Kac, *Lecture notes of 18.745 – introduction to lie algebras (fall 2010)*, `https://math.mit.edu/classes/18.745/classnotes.html`, 2010, Lecture notes, MIT.

[Nak] Hiraku Nakajima, *Lectures on hilbert schemes of points on surfaces*, University lecture series, American Mathematical Soc.

[Nak95] ———, *Heisenberg algebra and hilbert schemes of points on projective surfaces*, 1995.